Automating Homework Grading with Machine Learning

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Objective

The goal of this project is to automate the grading of student homework using machine learning. By analyzing various factors such as question type, subject, difficulty, correctness, and time taken, we aim to predict grades with high accuracy.

Dataset

The dataset used for this project contains features such as Question Type, Subject, Difficulty, Answer Length, Time Taken, and Correctness. The target variable is the Grade. The dataset includes 500 entries and underwent cleaning to remove duplicates and ensure no missing values were present.

Data Cleaning Results

- **Missing Values:** None.
- **Duplicates Removed:** 0.

Preprocessing

Categorical features (Question Type, Subject, Difficulty) were encoded using One-Hot Encoding, while numerical features (Answer Length, Time Taken, Correctness) were scaled using Standard Scaler. This ensures that the data is appropriately formatted for the model.

Model Training

Linear Regression was selected for its simplicity and interpretability. The dataset was split into training (80%) and testing (20%) sets to train and evaluate the model.

Model Evaluation

The following metrics were used to evaluate the model:

- **Mean Squared Error (MSE):** 0.3241
- **Root Mean Squared Error (RMSE):** 0.5693
- **R^2 Score:** 0.9608 (indicating the model explains approximately 96% of the variance in grades).

Visualizations

The following plots were generated to analyze the model and dataset:

1. **Scatter Plot: Correctness vs Grade** - Highlights the correlation between correctness and grades.
2. **Predicted vs Actual Grades** - Demonstrates how well the model predictions align with actual grades.
3. **Distribution of Prediction Errors** - Shows the spread of errors in the model predictions.

Conclusion

The Linear Regression model performed well, achieving a high R^2 score of 0.9608. This demonstrates its ability to accurately predict grades based on the features provided. Further improvements could be achieved by exploring advanced models or adding more diverse features.